

# Effect of minor additions on structure and volatilization loss in simulated nuclear borosilicate glasses

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## Abstract

Structural and thermal properties are reported for a range of caesium oxide-containing alkali borosilicate glasses, of the form  $x\text{Cs}_2\text{O}(100 - x)\text{ZMW}$  ( $0 < x < 10$ ), where ZMW represents a variety of simulated base-glasses. Glass densities increase and glass transition temperatures decrease with increase in caesium oxide concentration. Mass-loss from the melt is found to depend on composition in the same manner as the fraction of silicon Q<sup>3</sup> units, resolved from <sup>29</sup>Si MAS NMR, and is related to the presence of danburite medium-range order units, resolved from <sup>11</sup>B MAS NMR. Volatilization is shown to occur even in the absence of caesium oxide and the mixed alkali borosilicate composition of the volatile species, evolved from the melt at high temperature, is independent of the starting composition of the glass.

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## 1. Introduction

The efficient processing, immobilization and long-term storage of high level nuclear waste (HLW) is currently one of the most significant problems associated with the use of nuclear power as a serious replacement for fossil fuel-based energy production [1,2]. Of the many possible immobilization methods, vitrification of the waste using a mixed-alkali borosilicate glass waste form (MW), of composition 10.29 mol% Li<sub>2</sub>O, 10.53 mol% Na<sub>2</sub>O, 18.57 mol% B<sub>2</sub>O<sub>3</sub>, 60.61 mol% SiO<sub>2</sub>, is currently the preferred technique for HLW in the UK [3–5]. A significant problem associated with the process of HLW vitrification is the evolution of a volatile species at the glass melt stage. Although the volatile species is known to be caesium-containing [6–10], its overall

composition and the structural conditions affecting its formation have still to be fully established. We have sought to identify those structural changes which occur in the solid glass when caesium oxide is added to a variety of simulated waste-forms and attempt to relate these to the measured volatilization losses from the melt.

Many studies have been made of borosilicate glasses using NMR, Raman spectroscopy and various thermal analysis techniques [11–28] including a study of caesium oxide-doped MW [29]. These have all focussed on changes in the structure of the borate and silicate glass networks, as a function of alkali content, which lead to changes in density and thermal behavior. Structural studies, using <sup>11</sup>B, <sup>29</sup>Si and <sup>17</sup>O magic angle spinning nuclear magnetic resonance (MAS NMR), have led to widely accepted models for both the borate and borosilicate systems [11–13]. The addition of a glass modifier to a borate glass network initially converts trigonal planar, three-coordinated boron in [BO<sub>3</sub>]

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units ( $B_3$ ) to tetrahedral, four-coordinated boron in  $[BO_4]^-$  units ( $B_4$ ). At higher modifier concentrations,  $B_4$  are replaced by  $B_3^-$ , i.e., trigonal planar borons, with one non-bridging oxygen (NBO) [13,30–36]. The trend in the fraction of four-coordinated boron ( $N_4$ ) differs between the borate and borosilicate glass systems, although both are non-linear in nature. The most accurate  $N_4$  model to date for the borosilicate glass system is that of Dell et al. [13], although recently  $^{11}B$  MAS NMR measurements have shown deviations from this model for  $R > 0.5$ , where  $R$  is the ratio of free alkali oxide to boron oxide content.  $^{29}Si$  MAS NMR of borosilicate glasses has enabled the identification of the  $Q^n$  fractions ( $n$  denoting the number of bridging oxygens). Most commonly these are observed as  $Q^4$ ,  $Q^3$  and  $Q^4(B)$  units, where (B) indicates 3Si + 1B or 2Si + 2B next nearest neighbors [37].  $Q^2$  sites are only possible when there is sufficient alkali oxide associated with the silicate network that the ratio of alkali oxide to silica exceeds 1:2. The number of non-bridging oxygens per silicon (NBO/Si), i.e.,  $Q^3$  fraction, was predicted

$$NBO/Si = Q^3(\text{predicted}) = \frac{2 \times (\Sigma \text{mol\% } M_2O - \Sigma \text{mol\% } X_2O_3 - N_4 \times \text{mol\% } B_2O_3)}{\text{mol\% } SiO_2} \quad (1)$$

In this work,  $M_2O$  refers to modifiers  $Li_2O$ ,  $Na_2O$ ,  $Cs_2O$  and  $MgO$ , while  $X_2O_3$  refers to intermediates  $Al_2O_3$  and  $La_2O_3$ . Eq. (1) assumes that the trivalent cations form species of the type  $[XO_n]^-$ , where  $n$  is effectively four directly bonded oxygens for  $X = Al$ , B and 6–7 (including non-bonded) oxygens for La. The use of Eq. (1) is based on the assumption that  $NBO/Si \equiv Q^3$  for these glasses. The Dell model for speciation in borosilicate glasses [13] predicts that only  $Q^4$  and  $Q^3$  are present for the value of  $K$  ( $= [SiO_2]/[B_2O_3] = 3.2$ ) and the ranges of  $R$  ( $= ([M_2O] - [Al_2O_3, La_2O_3] + [MgO])/[B_2O_3]$ ) covered by these glass compositions. Previous work on related systems shows that this is true to even higher values of  $R$  than expected from the model [15].

Continuing from our previous study on  $Cs_2O$  additions to MW [29] we report the effect, on structure and volatilization, of the introduction of other oxide components of significance. These are  $Al_2O_3$ ,  $La_2O_3$  and  $MgO$ , in quantities which reflect their occurrence in HLW, outlined in Section 2.1 and Table 1. In the previous study [29], it was found that there was a correlation between mass-loss from

the melt and the fraction of  $Q^3$  units resolved from  $^{29}Si$  MAS NMR. The latter depends on the structural role of the additional oxide, i.e.,  $Al_2O_3$ ,  $La_2O_3$  or  $MgO$ , and how it affects the distribution of modifier cations between the borate and silicate network polyhedra.  $^{11}B$  and  $^{29}Si$  MAS NMR have been used in this current study to determine the speciation of the borate and silicate glass networks and their relationship with caesium volatility, from mass-loss measurements.

## 2. Experimental

### 2.1. Sample preparation

Four mixed-alkali borosilicate glass systems; Cs(AlMW), Cs(LaMW), Cs(MgMW) and Cs(Mg'MW), were made at compositions:  $xCs_2O(100 - x)ZMW$ , ( $0 < x < 10$ ) where ZMW represents a variety of simulated base-glasses to which  $Cs_2O$  is added (Table 1). These four base-glasses are modifications of the base-glass composition (MW) used by BNFL Ltd. for HLW vitrification, with a further oxide (Z) added, in a quantity realistic to that found in HLW. These additional oxides (and their initial molar quantities,  $y$ , before the addition of caesium oxide) are:  $Al_2O_3$  (0.95 mol%),  $La_2O_3$  (1.68 mol%) and  $MgO$  (2.55 and 10.2 mol%). MW is scaled down proportionally in the form  $y[Al_2O_3, La_2O_3, MgO](100 - y)MW$  to form the new base-glasses to which caesium oxide is then added.

For each mixed-alkali borosilicate system, 100 g batches were made with the appropriate reagent grade lithium carbonate (99.9%), sodium carbonate (99.95%), caesium carbonate (99.99%), aluminium oxide (99.9%), lanthanum oxide (99.99%), magnesium oxide (99.9%), sodium tetraborate (99.5%) and Wacomsil<sup>®</sup> quartz (99.9%) with 0.1 mol% iron (III) oxide (99.5%) added to reduce  $^{29}Si$  MAS NMR  $T_1$  relaxation times. Samples were mixed on rollers for 24 h before being transferred to 90% Pt/10% Rh crucibles and heated to between 1350 and 1400 °C (depending on composition) for 20 min before being cast into de-ionized water to form a frit. Samples were then re-melted at the same temperature and cast. Samples were analyzed by X-ray diffraction to confirm their amorphous character.

For use in volatilization measurements caesium-containing reedmergnerite and danburite samples were made to the compositions outlined in Table 2, using the appropriate

Table 1  
Starting compositions of the borosilicate glass systems

| System    | SiO <sub>2</sub> (mol%) | B <sub>2</sub> O <sub>3</sub> (mol%) | Na <sub>2</sub> O (mol%) | Li <sub>2</sub> O (mol%) | Al <sub>2</sub> O <sub>3</sub> , La <sub>2</sub> O <sub>3</sub> , MgO (mol%) | Fe <sub>2</sub> O <sub>3</sub> (mol%) | R    |
|-----------|-------------------------|--------------------------------------|--------------------------|--------------------------|------------------------------------------------------------------------------|---------------------------------------|------|
| MW        | 60.61                   | 18.57                                | 10.53                    | 10.29                    | –                                                                            | 0.1                                   | 1.12 |
| Cs(AlMW)  | 60.03                   | 18.39                                | 10.43                    | 10.19                    | 0.95                                                                         | 0.1                                   | 1.07 |
| Cs(LaMW)  | 59.59                   | 18.26                                | 10.35                    | 10.12                    | 1.68                                                                         | 0.1                                   | 1.03 |
| Cs(MgMW)  | 59.06                   | 18.10                                | 10.26                    | 10.03                    | 2.55                                                                         | 0.1                                   | 1.26 |
| Cs(Mg'MW) | 54.43                   | 16.68                                | 9.46                     | 9.24                     | 10.20                                                                        | 0.1                                   | 1.73 |

N.B.  $K = 3.2$  for all compositions.

Table 2  
Starting compositions for the danburite and reedmergnerite glass systems

| System         | Cs <sub>2</sub> O<br>(mol%) | B <sub>2</sub> O <sub>3</sub><br>(mol%) | SiO <sub>2</sub><br>(mol%) | Fe <sub>2</sub> O <sub>3</sub><br>(mol%) | R    | K    |
|----------------|-----------------------------|-----------------------------------------|----------------------------|------------------------------------------|------|------|
| Danburite      | 25.00                       | 25.00                                   | 50.00                      | 0.10                                     | 1.00 | 2.00 |
| Reedmergnerite | 12.50                       | 12.50                                   | 75.50                      | 0.10                                     | 1.00 | 6.00 |

carbonates and oxides listed above. Batches (20 g) were made of each, with powders being melted to between 1500 and 1550 °C, depending on composition, for 20 min before being quenched.

## 2.2. Density

Density measurements were carried out at room temperature, using the Archimedes method, with glycerol being used as a displacement fluid to avoid moisture attack on the high alkali-content glasses.

## 2.3. DTA

Differential thermal analysis (DTA) was carried out using a Perkin Elmer Diamond TG/DTA system, using an air flow of 200 ml/min as a purge gas. Typically 20–40 mg of sample and an equal mass of quartz powder reference were used in each experiment, with the sample being heated at 10 °C/min to 1000 °C and cooled, at the same rate, to room temperature.

## 2.4. Volatilization mass-loss

Mass-loss measurements were made by heating approximately 0.2 g of sample in a 90% Pt/10% Rh crucible at 10 °C/min to 1000 °C and held at temperature for 4 h, before being cooled at the same rate, to room temperature. Repeated measurements were made for each sample to obtain an average value for the mass-loss.

## 2.5. <sup>11</sup>B MAS NMR

<sup>11</sup>B MAS NMR spectra were obtained on a Varian/Chemagnetics Infinity 600 NMR spectrometer, operating at 192.4 MHz with a Chemagnetics MAS 4 mm probe spinning at 15 kHz. For each sample ~1000 acquisitions were taken with a pulse delay of 1 s, which was sufficient to enable relaxation, and a 0.7 μs pulse width (*B*<sub>1</sub> ~ 60 kHz) and acquisition delay of 8 μs.

Samples were referenced against BPO<sub>4</sub> solid taken as 3.3 ppm with respect to the primary reference Et<sub>2</sub>O:BF<sub>3</sub> at 0 ppm. Peak fits were made using the DMfit2002 NMR program [38].

## 2.6. <sup>29</sup>Si MAS NMR

<sup>29</sup>Si MAS NMR was carried out using a CMX Infinity 360 NMR spectrometer, operating at 71.43 MHz with a

6 mm Otsuka probe spinning at ~6 kHz. For each sample ~3000 acquisitions were taken with a 2 μs pulse width (30° tip angle), 5 s pulse delay, an acquisition delay of 50 μs and a line broadening of 50 Hz. All samples were referenced to tetramethylsilane (TMS) at 0 ppm.

## 2.7. EDX

A JEOL 6100 Scanning Electron Microscope with a 10 kV accelerating voltage was used for all EDX measurements. Measurements were taken on three alumina substrates that contained a sufficient deposit of the volatile species condensate, as well as one ‘blank’ alumina substrate with no deposit of material.

# 3. Results

## 3.1. Physical and thermal properties

The samples were amorphous within the limits of detection by XRD. The results from the density (including molar volume values) and DTA measurements are shown together in Figs. 1 and 2. Molar volume values (*V*<sub>m</sub>) were calculated from *V*<sub>m</sub> = *M*/*ρ*, where *M* is the molar mass of the glass in grams and *ρ* is the density in g/cm<sup>3</sup>. Apart from the glass transition and melt temperatures the DTA traces are featureless, reflecting the stability of all these glasses.

## 3.2. <sup>11</sup>B MAS NMR

The <sup>11</sup>B MAS NMR spectra show a large, relatively narrow peak from [BO<sub>4</sub>] at ~0 ppm and a small broader peak from [BO<sub>3</sub>] at ~13 ppm, with Fig. 3a being representative of the <sup>11</sup>B MAS NMR spectra for all four systems. Previous <sup>11</sup>B MAS NMR studies on borosilicate glasses [14,29] fitted a single Gaussian–Lorentzian lineshape to the [BO<sub>4</sub>]

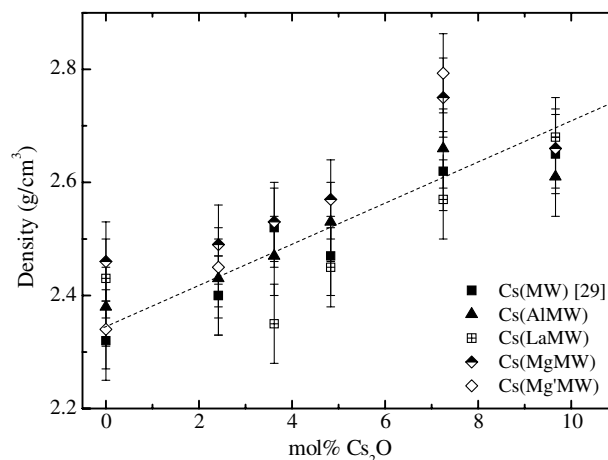


Fig. 1a. Density measurements for the four borosilicate glass systems, compared against previous work [29].

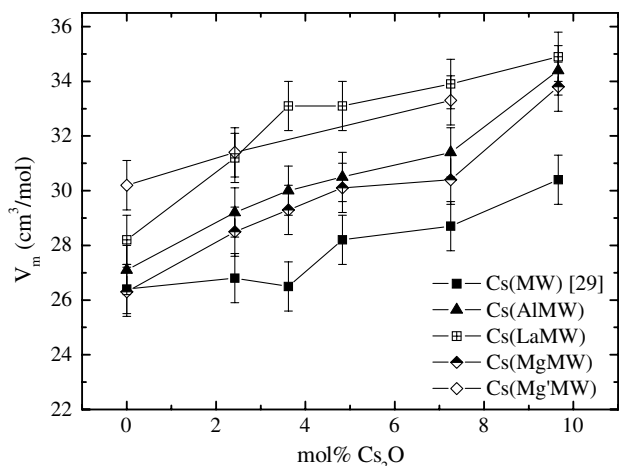


Fig. 1b. Molar volume values for the four borosilicate glass systems, compared against previous work [29].

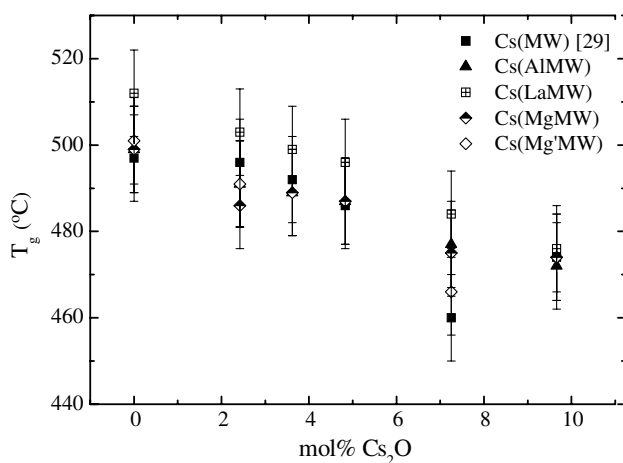


Fig. 2. Glass transition temperatures for the four borosilicate glass systems, compared against previous work [29].

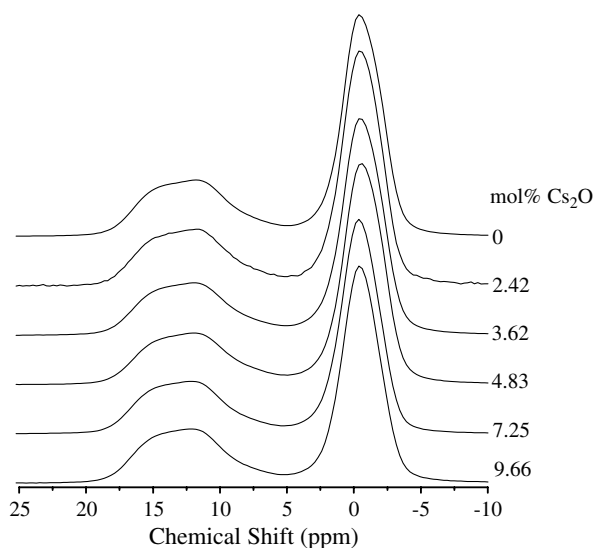


Fig. 3a.  $^{11}\text{B}$  MAS NMR spectra for the Cs(LaMW) system, typical for all of the borosilicate systems.

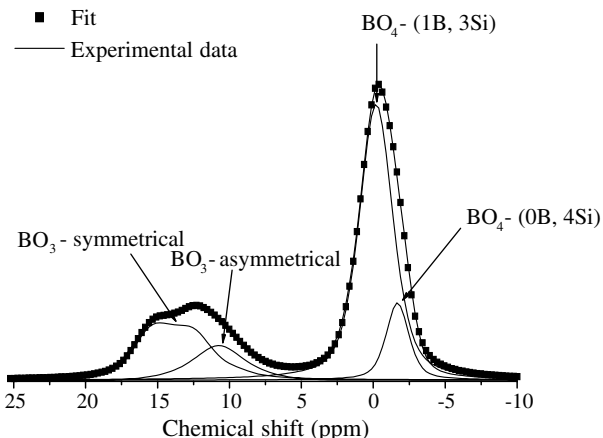


Fig. 3b. A typical  $^{11}\text{B}$  MAS NMR spectrum and fit.

region of the spectrum. However,  $^{11}\text{B}$  MQMAS NMR work by Du and Stebbins [20,39,40] demonstrated the presence of three distinct  $[\text{BO}_4]$  sites, with next nearest neighbors  $4\text{Si}$ ,  $1\text{B}_4 + 3\text{Si}$  and  $2\text{B}_4 + 2\text{Si}$ . The first two sites, with  $4\text{Si}$  and  $1\text{B}_4 + 3\text{Si}$  next nearest neighbors, have been assigned to the danburite ( $-0.6$  ppm) and reedmergnerite ( $-1.7$  ppm) medium-range order structures.

Spectra were therefore fitted with four peaks; two quadrupolar lineshapes, one for symmetric and one for asymmetric  $[\text{BO}_3]$  sites and two Gaussian–Lorentzian lineshapes representing the two  $[\text{BO}_4]$  sites (a good approximation where the quadrupole coupling constant is very small). Fits were judged to be unrealistic if the resolved half-width for the smaller peak (at  $\sim -1.5$  ppm) was greater than 2 ppm. A typical fit is shown in Fig. 3b. The inclusion of a third  $\text{B}_4$  site, with a  $2\text{B} + 2\text{Si}$  next nearest neighbor environment, was not required since  $K$  is sufficiently high ( $> 2$ ) that formation of these units would not occur, based on the observations of Du and Stebbins [39]. The ratios of the two resolved  $\text{B}_4$  peaks, for the four systems, are shown in Figs. 4a and b.

### 3.3. $^{29}\text{Si}$ MAS NMR and Mass-Loss Measurements

A typical set of  $^{29}\text{Si}$  MAS NMR spectra for the systems studied are shown in Fig. 5a. For the borosilicate glasses, the  $^{29}\text{Si}$  MAS NMR spectra were fitted in most cases with two Gaussian lines; one  $\text{Q}^3$  peak and one  $\text{Q}^4(\text{B})$  peak, a silicon with  $3\text{Si} + 1\text{B}$  next nearest neighbors (Fig. 5b). Contributions from  $\text{Q}^2$  units were negligible for all but typically the highest  $\text{Cs}_2\text{O}$  contents and the high MgO content system Cs(Mg' MW).

Resolved  $\text{Q}^3$  fractions for all four systems were observed to increase as a function of caesium oxide content. For all but the Cs(Mg' MW) system, resolved  $\text{Q}^3$  values were consistent within error ( $\pm 10\%$ ) with the predicted NBO/Si values calculated using Eq. (1). Volatilization measurements for the four mixed-alkali borosilicate systems gave mass-losses between 0% and 40%, with the trend in mass-loss

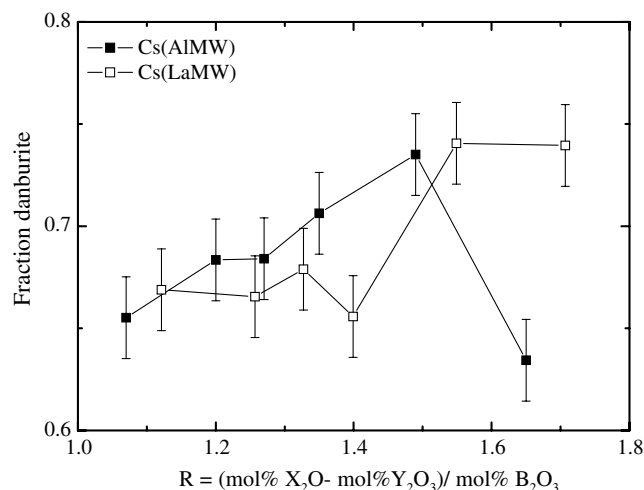


Fig. 4a. Resolved fraction of danburite units under the  $[\text{BO}_4]$  region of the  $^{11}\text{B}$  MAS NMR spectrum, for the Cs(AlMW) and Cs(LaMW) glass systems. Line is a guide to the eye [37].

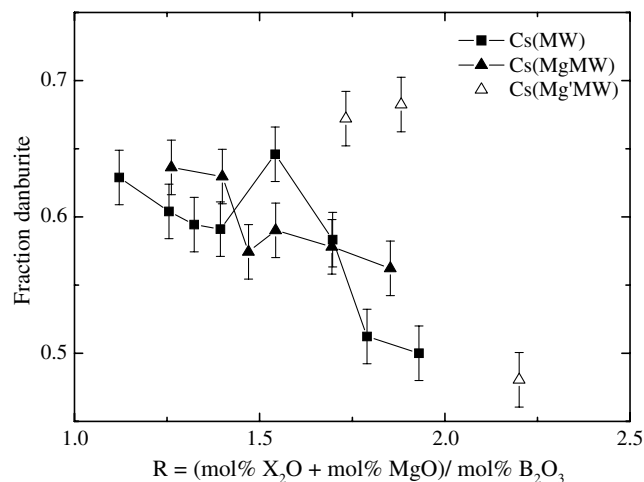


Fig. 4b. Resolved fraction of danburite units under the  $[\text{BO}_4]$  region of the  $^{11}\text{B}$  MAS NMR spectrum, for the groups I and II containing glass systems, including the Cs(MW) system from a previous study. Line is a guide to the eye [37].

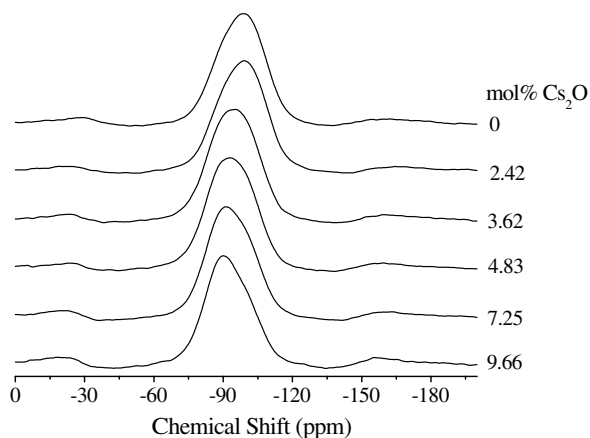


Fig. 5a.  $^{29}\text{Si}$  MAS NMR spectra for the Cs(AlMW) glass system. These are typical for all of the systems studied in this work.

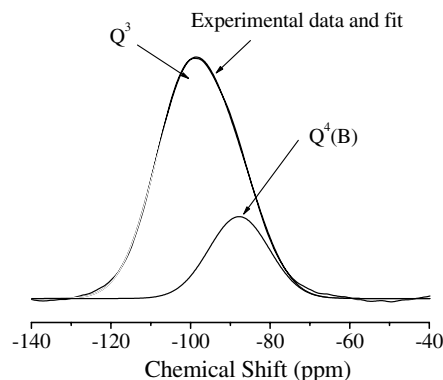


Fig. 5b. A typical  $^{29}\text{Si}$  MAS NMR fit for one of the mixed alkali borosilicate glasses.

being equivalent to that for the resolved  $\text{Q}^3$  fractions (Figs. 6 and 7).

EDX measurements of the volatile species (Fig. 8), from mass-loss measurements where a significant quantity of volatile species had been collected, showed the composition to contain caesium, sodium, boron and silicon. Light elements such as boron and lithium were not quantifiably detectable using EDX. The presence of aluminium, magnesium and calcium in the EDX spectra was due to the alumina substrate on which the volatile species were collected.

## 4. Discussion

### 4.1. Physical and thermal properties

The linear increase in density (Fig. 1a) is due to the dominating caesium oxide density ( $4.25 \text{ g/cm}^3$  compared to  $2.27 \text{ g/cm}^3$   $\text{Na}_2\text{O}$ ,  $2.01 \text{ g/cm}^3$   $\text{Li}_2\text{O}$ ,  $3.58 \text{ g/cm}^3$   $\text{MgO}$ ,  $3.97 \text{ g/cm}^3$   $\text{Al}_2\text{O}_3$ ,  $6.51 \text{ g/cm}^3$   $\text{La}_2\text{O}_3$ ,  $2.65 \text{ g/cm}^3$   $\text{SiO}_2$  and  $1.83 \text{ g/cm}^3$   $v\text{-B}_2\text{O}_3$ ). Although both aluminium oxide and lanthanum oxide possess relatively large densities, their quantities in the respective systems are relatively small (0.95 and 1.68 mol%, respectively) compared to the quantity of caesium oxide added. The molar volume ( $V_m$ ) also increases linearly with caesium oxide content (Fig. 1b), with the glass having to expand to accommodate the large  $\text{Cs}^+$  ions in the limited size voids in the network. Even in the absence of caesium oxide additions, all the systems show larger values of  $V_m$  than for the original MW base-glass [29] with the addition of  $\text{Al}_2\text{O}_3$ ,  $\text{La}_2\text{O}_3$  and  $\text{MgO}$  increasing  $V_m$  due to their large M–O distances (1.76 Å, 2.28 Å and 1.95 Å, respectively), compared to B–O (1.48 Å) and Si–O (1.62 Å). A greater change is seen initially in the Cs(Mg'MW) system, because of the significantly larger MgO content (Table 1).

Similarly, the decrease in  $T_g$  (Fig. 2) is consistent with the increase in caesium oxide in the system. It is presumed that the determining factor for this decrease is the smaller heat of formation for  $\text{Cs}_2\text{O}$  ( $-345.8 \text{ kJ/mol}$ ) compared to

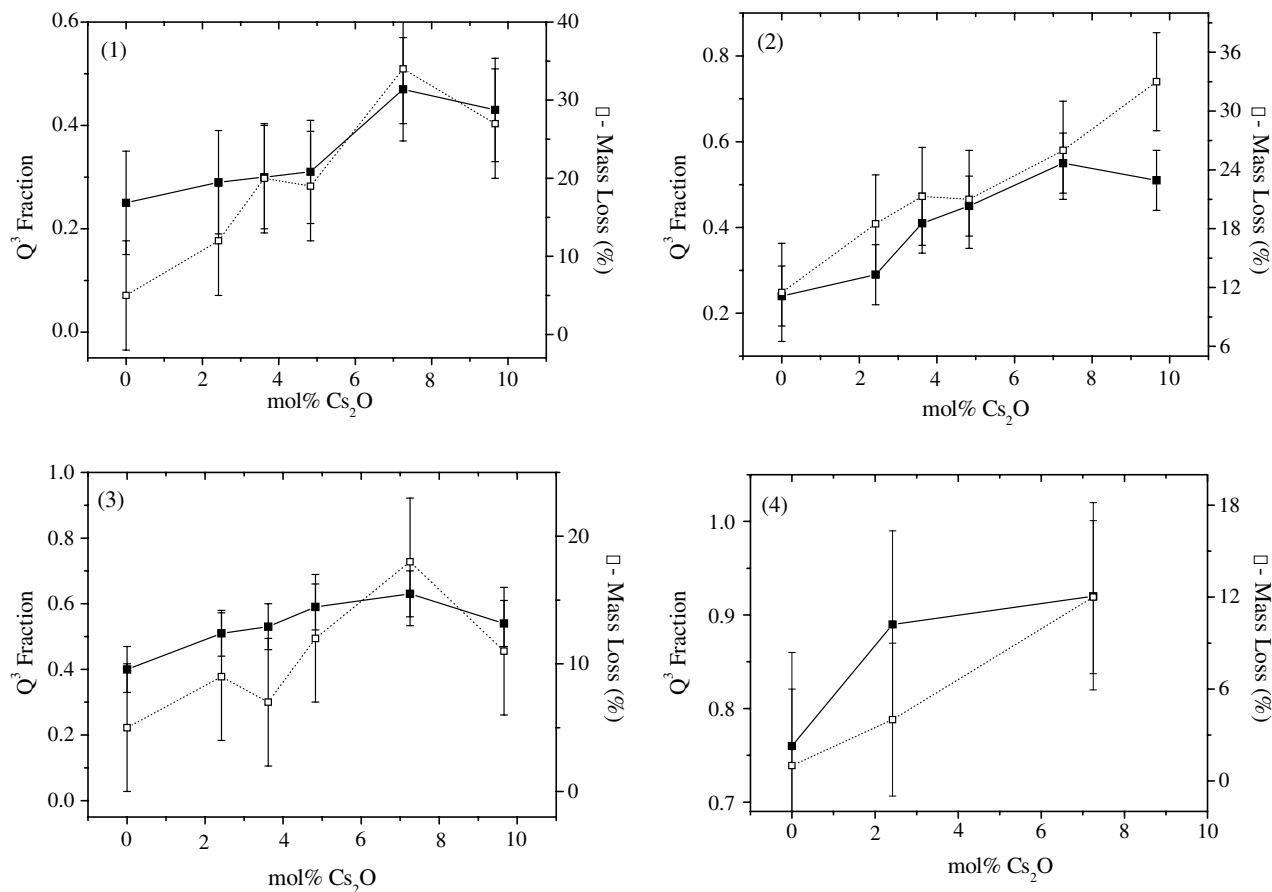


Fig. 6. Resolved  $Q^3$  fraction compared against mass loss (%) measurements for the Cs(AlMW) (1), Cs(LaMW) (2), Cs(MgMW) (3) and Cs(Mg' MW) (4) systems.

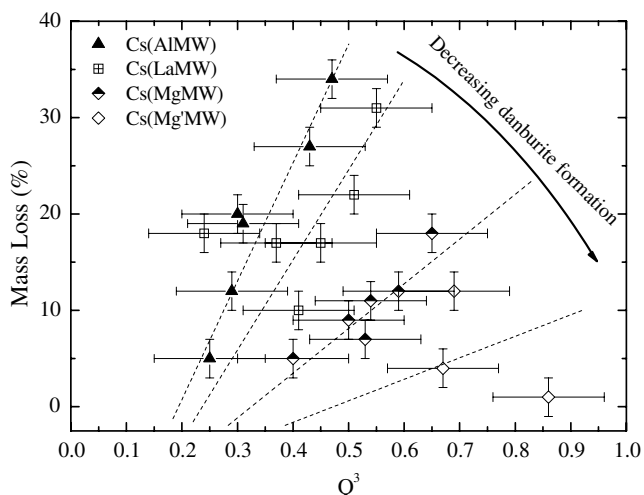


Fig. 7. Mass-loss as a function of resolved  $Q^3$  fraction for the four glass systems. Lines are a guide to the eye.

the other oxides present in all the glasses ( $-414$  kJ/mol  $Na_2O$ ,  $-597.9$  kJ/mol  $Li_2O$ ,  $-602.2$  kJ/mol  $MgO$ ,  $-1675.7$  kJ/mol  $Al_2O_3$ , and  $-1916.3$  kJ/mol  $La_2O_3$ ) indicating weaker bonds. For identical caesium oxide

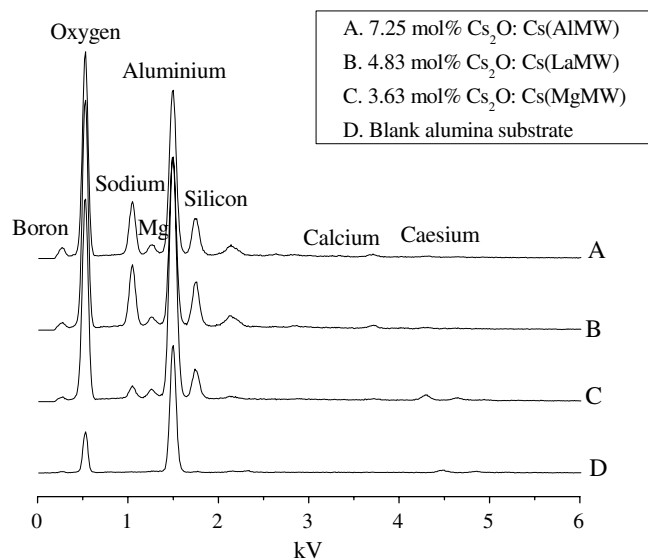


Fig. 8. EDX spectra of three selected deposits from volatilization mass-loss experiments. (a) 7.25 mol%  $Cs_2O$ , Cs(AlMW), (b) 4.83 mol%  $Cs_2O$  Cs(LaMW), (c) 3.63 mol%  $Cs_2O$  Cs(MgMW) and (d) blank alumina substrate.

contents, the greater values of  $T_g$  for the Cs(LaMW) compared to the remaining four systems, is also consistent with



the greater heat of formation for  $\text{La}_2\text{O}_3$ , relative to the other oxides.

#### 4.2. NMR and volatilization measurements

The fraction of danburite units under the  $[\text{BO}_4]$  region of the  $^{11}\text{B}$  MAS NMR spectrum (Fig. 4a and b) increases from approximately 0.65–0.75, for the Cs(AlMW) and Cs(LaMW) systems, and decreases from 0.65 to 0.50, for the Cs(MW) [29] and MgO-containing borosilicate systems. For the aluminium oxide- and lanthanum oxide-containing systems, the increase in the formation of danburite units, over the formation of reedmergnerite groups, is interesting due to the energetically unfavorable  $\text{B}_4\text{--B}_4$  bond in the danburite unit. Danburite formation has also been observed by Du and Stebbins [39] in a range of sodium-borosilicate glasses, with various  $K$  values ( $= 0.5, 2$  and  $4$ ), with the fraction of danburite units increasing from 0 to 0.4, as a function of  $R$  ( $0 \leq R \leq 0.7$ ). In this study, systems with increasing danburite contents, Cs(AlMW) and Cs(LaMW), are also the most volatile of the four systems, suggesting the less stable  $\text{B}_4\text{--B}_4$  bond contributes, together with  $\text{Q}^3$  formation, to increased volatility (Figs. 6 and 7). To confirm this observation, volatilization measurements were performed on the glasses of the danburite and reedmergnerite compositions. The danburite glass lost  $12 \pm 2\%$  mass and the reedmergnerite lost  $4 \pm 2\%$ . The occurrence of these two superstructural units in glasses is discussed in more depth in [37].

As discussed previously [29], mass-loss increases as a function of caesium oxide content in a manner comparable to the observed trend in resolved  $\text{Q}^3$  values from  $^{29}\text{Si}$  MAS NMR measurements. The question of whether the presence of caesium oxide is essential for volatilization to occur, originally discussed in [29], is highlighted again in this work; mass-loss measurements on those glasses with 0 mol%  $\text{Cs}_2\text{O}$ , show volatilization still occurs (Figs. 6 and 7) but increases almost linearly with further addition of  $\text{Cs}_2\text{O}$ . This appears to be a consequence of the accompanying increase in  $\text{Q}^3$  silicon species, and therefore, might equally be produced by equivalent additions of other modifiers. However, the mass-loss and the  $\text{Q}^3$  dependence do differ between systems, and therefore, the fraction of  $\text{Q}^3$  cannot be the only factor influencing mass-loss.

Compositional measurements of the volatile species, although limited to qualitative EDX measurements, because of the presence of boron and lithium (which cannot be measured quantitatively with EDX), show that the oxide composition of the volatile species is identical given the presence of either aluminium-, lanthanum- or magnesium-oxide in the starting composition of the borosilicate glasses. Results (Fig. 8) show the volatile species to contain caesium, sodium, boron, silicon and probably lithium, suggesting the composition to be mixed-alkali borosilicate in form. The studies carried out by Asano et al. [7,9,10] and Bonnell et al. [8] on vaporization from

caesium-containing borosilicate glasses, found the composition of the vapour to be alkali borate in form, devoid of any silicon.

#### 5. Conclusions

Density and thermal analysis measurements on a range of caesium oxide-containing mixed alkali borosilicate glasses have shown that the presence of caesium oxide is the dominating factor in their variation, with minimal change due to addition of the small quantities of aluminium oxide, lanthanum oxide or magnesium oxide, which are typical of their abundance in vitrified HLW. Mass-loss measurements show that volatilization from the melt occurs in the absence of caesium oxide in all of the base compositions, but increases linearly with increased addition of  $\text{Cs}_2\text{O}$ , with values being greater for the Cs(LaMW) and Cs(AlMW) systems.  $^{29}\text{Si}$  MAS NMR measurements show that the  $\text{Q}^3$  fraction also increases with  $\text{Cs}_2\text{O}$  addition, suggesting a relationship between  $\text{Q}^3$  fraction and volatility.  $^{11}\text{B}$  MAS NMR measurements of the medium-range order reedmergnerite and danburite structures present in all compositions, confirm that volatility is more pronounced in aluminium oxide- and lanthanum oxide-containing glasses, where there is an increasing presence of danburite units, containing energetically less favorable  $\text{B}_4\text{--B}_4$  bonds and this is confirmed by volatilization measurements on caesium-containing danburite and reedmergnerite glasses. The composition of the volatile species was found to be of a mixed-alkali borosilicate form, whereas previous results in similar studies have found it to be alkali borate [7–10].

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